

Solve only 4 , 5 no ques for assignment 4

1. Consider the following instruction mix given below. In a single-cycle Datapath scenario, answer the following questions:

R-type	I-type (non-lb)	Load	Store	Branch	Jump
15%	20%	20%	10%	25%	10%

- For **what** fraction of all instruction register write signals are enabled?
- For **what** fraction of instructions does the final output come directly from the ALU?
- For **what** fraction of all instruction data needs to perform sign extension?

2.

Design a single-cycle Datapath for the following instruction. Your diagram should have labeling of bit lengths and data flow along with the control signals. Please note that you have to design a Datapath that can only execute the instructions below. **Designing a general Datapath will have a severe penalty.**

BNE X₃, X₄, Else

3.

Explain the following statements:

- BNE can generate a control hazard.
- A data hazard can occur between two arithmetic instructions.
- LD data hazard needs a stall and then forwarding to overcome it.

4.

In the table below, we examine how pipelining affects the clock cycle time of the processor. This problem assumes that individual stages of the datapath have the following latencies:

IF	ID	EX	MEM	WB
280ps	200ps	210ps	280ps	200ps

- What** is the clock cycle time in a pipelined and non-pipelined processor?
- What** is the total time required for an instruction set having 5 instructions? Consider there are two data hazards, and each needs 2 stalls to overcome the hazards.

5.

ADD X₂₁, X₂₅, X₂₅

LD X₂₁, 24(X₂₅)

SD X₂₁, 24(X₂₅)

For the above instruction set running in a pipelined processor,

- Solve** the data hazard using only Forwarding.
- Find** the total number of clock cycles required to run the instruction set.